Adrian Ka-Wai Chung

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Academic appointments

- Oct 2026 -
  College Assistant Professor in Applied Mathematics, St John's College, Cambridge
- Oct 2025 - Sep 2026
  Herchel Smith Research Fellow in the Department of Applied Mathematics and Theoretical Physics (DAMTP), University of Cambridge
- Sep 2025 - Sep 2026
  College Research Associate and Supervisor, St John's College, Cambridge
- Nov 2022 - Sep 2025
  Postdoctoral Research Associate in the group of Prof. Nicolás Yunes at the University of Illinois Urbana-Champaign (UIUC)
- June 2015 - August 2015
  Summer research exchange student in Prof. Gregory Tucker's group at Brown University, specializing in analyzing data collected by the Spitzer telescope for exoplanet search.

Education

- 2019-2022 (degree conferred on 1st January 2023)
  Doctor of Philosophy in Physics
  Theoretical Particle Physics and Cosmology Group, Department of Physics, King's College London
  Supervisor: Prof. Mairi Sakellariadou
- 2017-2019 (degree conferred in 2020)
  Master of Philosophy in Physics
  Department of Physics, The Chinese University of Hong Kong
  Supervisor: Prof. Tjonnie G.F. Li
- 2013-2017
  Bachelor of Science in Physics, First Class Honours
  Department of Physics, The Chinese University of Hong Kong
  Supervisor: Prof. Tjonnie G.F. Li

Collaborations and research leadership

- 2026 -
  Core Member of Laser Interferometer Space Antenna (LISA) Consortium
- 2026 -
  Member of the Simons Collaboration on Black Holes and Strong Gravity (SimonsC-BHSG, https://simonscbhsg.com/)
- 2024 - 2025
  Member of the National Science Foundation - Simons Artificial Intelligence Institute for the Sky (SkAI)
- 2022 - 2025
  Member of the Laser Interferometer Space Antenna (LISA) consortium
- 2022 - 2025
  Member of the Illinois Center for Advanced Studies of the Universe (ICASU)
- 2022 -
  Member of the American Physical Society (APS)
- 2016 - 2022
  Member of the Laser Interferometer Gravitational-wave Observatory (LIGO) Scientific Collaboration. I was an active member of the Testing General Relativity (TGR) working group. I led a collaboration-wide

TGR analysis whose results are included in several LIGO-Virgo-KAGRA collaboration publications (see publication list).

Research funding and major awards

Through my research career, I have secured a total of more than 650k USD in funding, including scholarships, computation grants[1], and awards.

1. Herchel Smith Research Fellowship, University of Cambridge, 15000 GBP research allowance (in addition to salary) per annum, 2025 - 2026
2. College Research Associateship, St John's College, University of Cambridge, 2025 - 2026
3. (Declined) Post-Doctoral By-Fellowship, Churchill College, University of Cambridge, 1000 GBP research allowance per annum, 2025 - 2028
4. (Declined) Rokos Postdoctoral Research Associateship, Queens' College, University of Cambridge, 1000 GBP research allowance per annum, 2025 - 2027
5. Computational grant, "PHY240142: Simulating black-hole pulsations in modified gravity", the National Science Foundation (NSF) through the Office of Advanced Cyberinfrastructure (ACCESS), PI, 400k USD, 2024
6. Hong Kong Scholarship for Excellence Scheme ($>$ 1 million HKD $>$ 130k EUR $\sim$ 150k USD in total), by the HKSAR Government, 2019 - 2022
7. Best student presentation prize in the quantum gravity parallel session at the 30th Texas Symposium on Relativistic Astrophysics, 2019
8. Teaching-Assistant Award, by the Department of Physics of the Chinese University of Hong Kong, 2018 and 2019
9. Reaching Out Award (10 k HKD), by the HKSAR Government, 2018
10. CN Yang Scholarship (≤ 10 k HKD), by Prof. Chen-Ning Yang and the Department of Physics of the Chinese University of Hong Kong, 2014, 2015 and 2017
11. Professor and Mrs Yau Wa Chan Scholarship (10k HKD), by the Department of Physics of the Chinese University of Hong Kong, 2016
12. Dean's Honour List, by the Faculty of Science of the Chinese University of Hong Kong, 2014, 2015 and 2016
13. Department/Programme Scholarship (5k HKD per award), by the Shaw College of the Chinese University of Hong Kong, 2014, 2015 and 2016
14. College Head List, by the Shaw College of the Chinese University of Hong Kong, 2014, 2015 and 2016
15. Talent Development Scholarship (10k HKD), by the HKSAR Government, 2015

Research supervision and mentoring

I have mentored 15 junior researchers.

1. **Carolin Kohl** — second-year PhD student at DAMTP, Cambridge — daily PhD supervisor — on modelling extreme-mass-ratio inspirals in beyond-GR theories and dark-sector environments, 2026–Present.
2. **Tomasz Kinowski** — second-year PhD student at the Institute of Astronomy, Cambridge — co-mentored with Prof. Christopher Moore — on gravitational-wave tests of General Relativity and the quasinormal modes of black holes in Stelle-type quadratic gravity, 2026–Present.
3. **Richard Dyer** — third-year PhD student at the Institute of Astronomy, Cambridge — co-mentored with Prof. Christopher Moore — on ringdown tests of General Relativity, 2026–Present.
4. **Leif Lui** — second-year Master's student at the Beijing Institute of Mathematical Sciences and Applications — mentored — on extreme-mass-ratio-inspiral probes of dark matter, 2026–Present.
5. **Stef J. B. Husken** — first-year PhD student in Physics at KU Leuven — co-mentored with Prof. Tjonnie G. F. Li — on computations of scalar quasinormal modes, 2025–Present.

[1] A computational grant refers to funding specifically allocated for purchasing computing time.

6. **Kelvin Ka-Ho Lam** — third-year PhD student at the University of Illinois Urbana-Champaign — co-mentored with Prof. Nicolás Yunes — on computing black-hole spacetimes and quasinormal modes in beyond General-Relativity gravity theories, 2024–2025.
7. **Bryce Cousins** — fourth-year PhD student at the University of Illinois Urbana-Champaign and NSF Graduate Research Fellow — co-mentored with Prof. Nicolás Yunes — on improving measurements of the Hubble constant by jointly analysing the binary-black-hole population and gravitational-wave background, 2024–2025. Bryce is also a recipient of the 2026 Jump Fellowship, supported by Jump Trading LLC (https://grad.illinois.edu/news/bryce-cousins-awarded-2026-jump-fellowship).
8. **Kristen Schumacher** — joint postdoctoral fellow at the California Institute of Technology and the University of Texas, Austin — co-mentored with Prof. Nicolás Yunes — on improving measurements of the Hubble constant by jointly analysing the binary-black-hole population and gravitational-wave background, 2024–2025.
9. **Pratik Wagle** — Leibniz Fellow at the Max Planck Institute for Gravitational Physics — co-mentored with Prof. Nicolás Yunes — on computing black-hole quasinormal modes in General Relativity, 2023.
10. **Joseph Gais** — global financial analyst at Nomura — co-mentored with Prof. Tjonnie G. F. Li — on dark-matter searches using black-hole ringdown signals, 2021.
11. **Mark Ho-Yeuk Cheung** — Fellow at the Institute for Advanced Study, Princeton — co-mentored with Prof. Tjonnie G. F. Li — on dark-matter searches using black-hole ringdown signals and ringdown spectroscopy of rotating black holes pierced by cosmic strings, 2019–2021. Mark received a Croucher Scholarship for Doctoral Study in 2022 and a Croucher Fellowship for Postdoctoral Research in 2025.
12. **Levi Wing-Hei Poon** — PhD student in Mathematics at the University of Illinois Urbana-Champaign — co-mentored with Prof. Tjonnie G. F. Li — on ringdown spectroscopy of rotating black holes pierced by cosmic strings, 2019. Levi received a Croucher Scholarship for Doctoral Study in 2021.
13. **Alan Tsz-Lok Lam** — postdoctoral research associate at Pennsylvania State University — co-mentored with Prof. Tjonnie G. F. Li — on numerical simulations of spherically symmetric core-collapse supernovae in Gauss–Bonnet theories, 2018.
14. **Alexander M. Tanaka** — PhD student in Biophysics at the University of Oxford — co-mentored with Prof. Tjonnie G. F. Li — on modifying the Teukolsky equation to incorporate generic dispersion of gravitational waves, 2018.
15. **Jacky Hoi-Tung Yip** — PhD student at the University of Wisconsin–Madison — co-mentored with Prof. Tjonnie G. F. Li — on developing numerical methods to compute black-hole quasinormal-mode frequencies in massive-gravity theories, 2018.

Teaching

1. Director of Studies in Mathematics, St John's College, University of Cambridge, From Lent Term 2027
2. Supervisor for Part II Statistical Physics, DAMTP, University of Cambridge, Lent Term, 2025-2026
3. Supervisor for Part IB Quantum Mechanics, St John's College, University of Cambridge, Michaelmas Term, 2025-2026
4. Teaching assistant for General Relativity, CUHK, Spring, 2018-2019
5. Teaching assistant for Electromagnetism I, CUHK, Fall, 2018-2019
6. Teaching assistant for Methods for Theoretical Physics I, CUHK, Spring, 2017-2018
7. Teaching assistant for Electromagnetism I, CUHK, Fall, 2017-2018

Selected publications

Below is a selection of my most significant publications. The full list is available at: https://inspirehep.net/authors/1879388?ui-citation-summary=true.

Short-author-list publications

1. Black Hole Fine Structure
   A.K.-W.Chung, Richard Dyer, Tomasz Kinowski, Christopher J. Moore
   https://arxiv.org/abs/2610.18989
2. Quasinormal modes of rotating black holes with ultralight scalar fields
   Leif Lui, **A.K.-W.Chung**
   https://arxiv.org/abs/2610.16054
3. Devoured by a Hairy Gargantua: Probing Massive Scalar Charges with Non-minimal Curvature Coupling with Extreme-Mass-Ratio Inspirals
   Leif Lui, **A.K.-W.Chung**, Alejandro Torres-Orjuela
   https://arxiv.org/abs/2608.28037
4. Modeling direct waves in binary black hole ringdowns
   Richard Dyer, **A.K.-W.Chung**, Christopher J. Moore
   https://arxiv.org/abs/2606.25021
5. Measuring a Black Hole's Area Immediately after Merger: A Direct-Wave Test of Hawking's Area Law
   A.K.-W.Chung, Kelvin Ka-Ho Lam, Anna Liu, Nicolás Yunes,
   https://arxiv.org/abs/2606.06592
6. Quadratic gravity corrections to scalar quasinormal modes of rapidly rotating black holes
   Stef J. B. Husken, Tom van der Steen, Simon Maenaut, Kelvin Ka-Ho Lam, Maxim D. Jockwer, **A.K.-W.Chung**, Thomas Hertog, Tjonnie G. F. Li, Nicolás Yunes,
   Phys. Rev. D. 114, 064072 (2026)
   https://journals.aps.org/prd/abstract/10.1103/jrnh-wsxw.
7. Black hole ringdown spectroscopy of curvature-coupled scalar and pseudoscalar fields
   A.K.-W.Chung, and Nicolás Yunes
   Phys. Rev. D. 114, L021505 (2026)
   https://journals.aps.org/prd/accepted/10.1103/2ctd-mgb6
8. Spacetime of rotating black holes surrounded by massive scalar charges
   A.K.-W.Chung
   Phys. Rev. D. 113, 095032 (2026)
   https://journals.aps.org/prd/abstract/10.1103/4qbd-f98r
9. Stochastic Siren: Astrophysical Gravitational-Wave Background Measurements of the Hubble Constant
   Bryce Cousins, Kristen Schumacher, **A.K.-W.Chung**, Thomas Callister, Colm Talbot, Daniel E. Holz, and Nicolás Yunes,
   Phys. Rev. Lett., 136, 101003 (2026)
   https://journals.aps.org/prl/abstract/10.1103/4lzh-bm7y
   This work is also reported by *Physics World* in
   https://physicsworld.com/a/gravitational-effects-could-shed-more-light-on-the-hubble-tension/
10. Spinning Black Holes in Modified Gravity via Spectral Methods
    Kelvin Ka-Ho Lam, **A.K.-W.Chung**, and Nicolás Yunes,
    Phys. Rev. Lett. 136, 021401 (2026)
    https://journals.aps.org/prl/abstract/10.1103/6z43-xdv7
11. Analytic and accurate approximate metrics for black holes with arbitrary rotation in beyond-Einstein gravity using spectral methods
    Kelvin Ka-Ho Lam, **A.K.-W.Chung**, and Nicolás Yunes,
    Phys. Rev. D. 113, 024030 (2026)
    https://journals.aps.org/prd/abstract/10.1103/txxb-z8bx
12. Quasi-normal mode frequencies and gravitational perturbations of black holes with any subextremal spin in modified gravity through METRICS: the dynamical Chern-Simons gravity case
    A.K.-W.Chung, Kelvin Ka-Ho Lam, and Nicolás Yunes,
    Phys. Rev. D. 111, 124052 (2025)
    https://journals.aps.org/prd/abstract/10.1103/g83n-rrlj

13. Bayesian search for massive scalar charge from LIGO-Virgo-KAGRA binaries
Yiqi Xie, **A.K.-W.Chung**, Thomas Sotiriou, and Nicolás Yunes,
Phys. Rev. Lett. 134, 191402 (2025)
https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.134.191402
14. Ringing out General Relativity: Quasi-normal mode frequencies for black holes of any spin in modified gravity
A.K.-W.Chung, and Nicolás Yunes
Phys. Rev. Lett. 133, 181401 (2024)
https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.133.181401
15. Quasi-normal mode frequencies and gravitational perturbations of black holes with any subextremal spin in modified gravity through METRICS: the scalar-Gauss-Bonnet gravity case
A.K.-W.Chung, and Nicolás Yunes,
Phys. Rev. D 110, 064019 (2024)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.110.064019
16. Spectral method for the gravitational perturbations of black holes: Kerr background case
A.K.-W.Chung, Pratik Wagle, and Nicolás Yunes,
Phys. Rev. D 109, 044072 (2024)
https://doi.org/10.1103/PhysRevD.109.044072
17. Untargeted Bayesian search of anisotropic gravitational-wave backgrounds through the analytical marginalization of the posterior
A.K.-W.Chung and Nicolás Yunes,
Phys. Rev. D 108, 043032 (2023)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.108.043032
18. Spectral method for the gravitational perturbations of black holes: Schwarzschild background case
A.K.-W.Chung, Pratik Wagle, and Nicolás Yunes,
Phys. Rev. D 107, 124032 (2023)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.107.124032
19. Targeted search for the kinematic dipole of the gravitational-wave background
A.K.-W.Chung, Alexander C. Jenkins, Joseph D. Romano, Mairi Sakellariadou,
Phys. Rev. D 106, 082005 (2022)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.106.082005
20. Searching for ultra-light bosons with supermassive black hole ringdown
A.K.-W.Chung, Joseph Gais, Mark Ho-Yeuk Cheung and Tjonnie G.F. Li,
Phys. Rev. D 104, 084028 (2021)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.104.084028
21. Lensing of gravitational waves as a novel probe of graviton mass
A.K.-W.Chung and Tjonnie G.F. Li,
Phys. Rev. D 104, 124060 (2021)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.104.124060
22. Upper limits on the temperature of inspiraling astrophysical black holes
A.K.-W.Chung and Sakellariadou, M.,
The European Physical Journal C 81, 592 (2021)
https://link.springer.com/article/10.1140/epjc/s10052-021-09391-3
23. Ringdown spectroscopy of rotating black holes pierced by cosmic strings
Mark Ho-Yeuk Cheung, Levi Wing-Hei Poon, **A.K.-W.Chung** and Tjonnie G.F. Li,
Journal of Cosmology and Astroparticle Physics 02(2021)040
https://iopscience.iop.org/article/10.1088/1475-7516/2021/02/040
24. Phenomenological inclusion of alternative dispersion relations to the Teukolsky equation and its application to bounding the graviton mass with gravitational-wave measurements
A.K.-W.Chung and Tjonnie G.F. Li,
Phys. Rev. D 99, 124023 (2019)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.99.124023

LIGO-Virgo-Collaboration publications with significant contributions

When I was a member of the LIGO Scientific Collaboration (LSC), I made significant contributions to the following LSC publications.

25. Tests of General Relativity with Binary Black Holes from the second LIGO-Virgo Gravitational-Wave Transient Catalog
Abbott, B. P. *et al.* (The LIGO Scientific Collaboration and the Virgo Collaboration),
Phys. Rev. D 103, 122002 (2021)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.103.122002
26. Tests of general relativity with the binary black hole signals from the LIGO-Virgo catalog GWTC-1
Abbott, B. P. *et al.* (The LIGO Scientific Collaboration and the Virgo Collaboration),
Phys. Rev. D 100, 104036 (2019)
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.100.104036

Review articles

I was invited to participate in writing the following review article.

27. Black hole spectroscopy: from theory to experiment
Emanuele Berti *et al.*
Class. Quantum Grav. 43 123001
https://iopscience.iop.org/article/10.1088/1361-6382/ae59e2

Popular science articles

28. Ringdown: the music by oscillating black holes
A.K.-W.Chung
invited article contribution to the Hong Kong Laureate Newsletter, October 2023
https://hklaureateforum.org/en/ringdown-the-music-by-oscillating-black-holes

Professional service

I have served as a referee for the following journals:

- Physical Review Letters
- Physical Review D
- The Astrophysical Journal
- The European Physical Journal C
- Classical and Quantum Gravity
- International Journal of Modern Physics D

Selected invited talks and conference attendance

I have given 16 invited seminars/lectures and 12 contributed talks. I was also invited to chair a session at different international conferences.

1. "Ringdown in Lisbon", Academia das Ciências of Lisbon, **invited participant**, Lisbon Portugal, 19th - 21st October 2026
2. *Black-hole METRICS, ringdown and fundamental physics*, **invited seminar** at CamGW meeting, Cambridge United Kingdom, 4th June 2026
3. *Measuring a Black Hole's Area Immediately after Merger: A Direct-Wave Test of Hawking's Area Law*, webinar presentation at the LIGO-Virgo-KAGRA Collaboration, 25th May 2026
4. Simons Collaboration on Black Holes and Strong Gravity Annual Meeting, **invited participant**, New York United States of America, 7th - 8th May 2026
5. *Black-hole METRICS, ringdown and fundamental physics*, Theoretical Astrophysics Including Relativity and Cosmology (TAPIR), California Institute of Technology, California United States of America, 3rd April 2026
6. American Physical Society Global Summit, **invited chair** of the session "Astrophysics with Gravitational Waves II", Colorado United States of America, 18th March 2026
7. *Probing quadratic gravity with black-hole ringdown gravitational waves measured by LIGO-Virgo-KAGRA detectors*, contributed presentation, American Physical Society Global Summit, Colorado United States of America, 18th March 2026

8. *Black-hole METRICS, ringdown and fundamental physics*, **invited seminar** at the Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge United Kingdom, 23rd January, 2026
9. *Black-hole METRICS, ringdown and fundamental physics*, **invited seminar** at University College London, London United Kingdom, 22nd January 2026
10. *Extracting Fundamental Physics through Black-Hole Gravitational-Wave Ringdown Signals*, contributed seminar, the Institute of Advanced Study of the Hong Kong University of Science and Technology, Hong Kong China, 13th November 2025
11. *Black-Hole Ringdown Spectroscopy as a Precision Probe of Gravity*, contributed seminar, the Chinese University of Hong Kong, Hong Kong China, 10th November 2025
12. *Listening to the Sounds of Black Holes*, **invited** predinner lecture, St John's College, Cambridge United Kingdom, 14th October 2025
13. 24th International Conference on General Relativity and Gravitation (GR24) and the 16th Edoardo Amaldi Conference on Gravitational Waves (Amaldi16), **invited chair** of the Session "Experimental tests of the nature of compact objects", Glasgow United Kingdom, 17th July 2025
14. *METRICS and its use in extracting fundamental physics from gravitational-wave ringdown signals*, contributed presentation, 24th International Conference on General Relativity and Gravitation (GR24) and the 16th Edoardo Amaldi Conference on Gravitational Waves (Amaldi16), Glasgow United Kingdom, 14th July 2025
15. *METRICS as a bridge between black-hole ringdown signals and fundamental physics*, **invited seminar**, Queen Mary University of London, London United Kingdom, 7th July 2025
16. **invited panelist** of the discussion panel on "Tests of gravity", 10th Physics and Astrophysics at the eXtreme Workshop and 3rd Cosmic Explorer Symposium, 30th June - 3rd July 2025
17. *Probing quadratic gravity with black-hole ringdown gravitational waves measured by LIGO-Virgo-KAGRA detectors*, **invited seminar**, KU Leuven, Leuven Belgium, 20th June 2025
18. *Gravitational-wave tests of general relativity with METRICS*, **invited seminar**, the University of Virginia, Charlottesville United States of America, 28th April 2025
19. *Computations of black-hole quasinormal modes in dynamical Chern-Simons gravity using METRICS*, contributed talk, American Physical Society Global Summit, California United States of America, 19th March 2025
20. *Gravitational-Wave Tests of General Relativity with METRICS*, contributed seminar, the Institute of Advanced Study of the Hong Kong University of Science and Technology, Hong Kong China, 11th February 2025
21. *Gravitational-wave tests of general relativity with METRICS*, contributed seminar, the Chinese University of Hong Kong, Hong Kong China, 24th January 2025
22. *Gravitational-wave tests of general relativity with METRICS*, contributed seminar, the University of Hong Kong, Hong Kong China, 22nd January 2025
23. *Gravitational-wave tests of general relativity with METRICS*, contributed seminar, the City University of Hong Kong, Hong Kong China, 21st January 2025
24. *Gravitational-wave tests of general relativity with METRICS*, contributed presentation, 2025 IAS Program on Fundamental Physics, the Institute of Advanced Study of the Hong Kong University of Science and Technology, Hong Kong China, 15th January 2025
25. **Invited visitor** to Prof. Yanbei Chen's group, Theoretical Astrophysics Including Relativity and Cosmology (TAPIR), California Institute of Technology, California United States of America, 11th - 16th November, 2024
26. *METRICS and its applications to extract physics through gravitational-wave detection*, **invited seminar**, Perimeter Institute, Waterloo Canada, 31st October 2024
27. *METRICS and its applications to extract physics through gravitational-wave detection*, **invited seminar**, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam Germany, 13th September 2024
28. *METRICS and its applications to extract physics through gravitational-wave detection*, **invited seminar**, KU Leuven, Leuven Belgium, 6th September 2024
29. Black Hole Inside and Out, **invited** participant, Niels Bohr Institute, Copenhagen Denmark, 26th - 31st August 2024

30. Ringdown Inside and Out, **invited** participant, Niels Bohr Institute, Copenhagen Denmark, 22nd - 24th August 2024
31. *METRICS and its applications to extract physics through gravitational-wave detection*, **invited seminar**, Niels Bohr Institute, Copenhagen Denmark, 20th August 2024
32. *Extracting fundamental physics through the detection of the ringdown phase by LISA*, contributed talk, The 15th International Laser Interferometer Space Antenna (LISA) Symposium, Dublin Ireland, 10th July 2024
33. *Black-hole ringdown phase in modified gravity using METRICS*, contributed talk, American Physical Society April Meeting, California United States of America, 4th April 2024
34. American Physical Society April Meeting, **invited chair** of the session "Tests of General Relativity", California United States of America, 3rd April 2024
35. *Rotating black-hole METRICS*, **invited seminar**, the Chinese University of Hong Kong, Hong Kong China, 26th February 2024
36. *Rotating black-hole METRICS*, contributed talk, YITP long-term workshop Gravity and Cosmology 2024, Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto Japan, 24th February, 2024
37. Selected participant and speaker for flash presentation, Hong Kong Laureate Forum, Hong Kong China, 12th - 18th November 2023
38. *Spectral method for computing gravitational quasinormal-mode frequencies of black hole*, **invited seminar**, Theoretical Astrophysics Including Relativity and Cosmology (TAPIR), California Institute of Technology, California United States of America, 18th October 2023
39. *Spectral method for computing gravitational quasinormal-mode frequencies of black hole*, contributed presentation, Amaldi15 Premier International Conference on Gravitational Waves (online), 17th July 2023
40. *Untargeted Bayesian search of anisotropic gravitational-wave backgrounds through the analytical marginalization of the posterior*, **invited** presentation at the call of the anisotropic gravitational-wave background working group of LIGO, 22nd May 2023
41. *Spectral method for computing gravitational quasinormal-mode frequencies of black hole*, contributed presentation, American Physical Society April Meeting, Minnesota United States of America, 15th April 2023
42. *Spectral method for computing gravitational quasinormal-mode frequencies of black hole*, **invited seminar**, the Department of Physics, The University of Wisconsin-Milwaukee, Wisconsin United States of America, 3rd February, 2023
43. *Searching for ultralight bosons with supermassive black hole ringdown*, **invited seminar**, the Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge United Kingdom, 13th July, 2022
44. *Searching for ultralight bosons with supermassive black hole ringdown*, (online) contributed talk, Workshop on Very Light Dark Matter 2021, 28th September 2021
45. *Searching for ultralight bosons with supermassive black hole ringdown*, (online) contributed talk, Gravitex 2021: International Conference on Gravitation, 9th August 2021
46. *Probing the purely ingoing nature of the black-hole event horizon*, contributed talk, the 30th Texas Symposium on Relativistic Astrophysics, The University of Portsmouth, Portsmouth United Kingdom, 15th - 20th December 2019.
47. *Probing the purely ingoing nature of the black-hole event horizon*, contributed talk, Gravitational Wave Physics and Astronomy Workshop, The University of Tokyo, Tokyo Japan, 16th October 2019
48. *A phenomenological inclusion of alternative dispersion relations to the Teukolsky equation and its application to bounding the graviton mass with gravitational-wave measurements*, contributed talk, Testing Gravity 2019, Simon Fraser University, Vancouver Canada, 24th January 2019
49. *Testing General Relativity Through Gravitational-Wave Detection*, **invited lecture**, 2019 TGWG workshop on gravitational wave data analysis, Taiwan China, 21st January 2019
50. *Constraining graviton mass with black-hole ringdown*, **invited seminar**, the University of Pisa, Pisa Italy, August 2018
51. Updating the LIGO-Virgo Collaboration on the residual analysis, LIGO-Virgo Collaboration Meeting, August 2018

52. *Constraining graviton mass with black-hole ringdown*, **invited seminar**, the Dutch National Institute for Subatomic Physics (Nikhef), Amsterdam Netherlands, August 2018
53. *Constraining Lorentz violation of gravitational waves with lensing*, Seminar in Workshop "Gravity and Cosmology 2018", Yukawa Institute of Theoretical Physics, Kyoto University, Kyoto Japan, 6th February 2018
54. Summer School of Gravitational Waves, invited participant, The International Centre for Theoretical Sciences (ICTS) of the Tata Institute of Fundamental Research, Bengalore India, 2016